Individual Contribution Statement

Name: Alycia Reji

Team: Silly Geese (Alycia Reji, Gracie Vivion, Shelby Howard, Daniz Mammadova)

Date: May 1, 2026

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1. Personal Contribution to Capstone Milestones

Milestone 1: Data Pipeline (Week 5)

Tasks completed:

- Cleaned lobbying disclosure data from Senate files (removed amendments and no-activity filings)
- Built CIK-GVKEY crosswalk to link lobbying and financial datasets
- Merged multiple data sources on firm identifiers and year to create final panel dataset
- Verified merge logic and checked for missing values and duplicate rows

Hours spent: 12 hours

Key deliverable: CIK-GVKEY crosswalk and merged datasets (merged_financials_lobbying.csv and balanced panel version)

Milestone 2: EDA Dashboard (Week 10)

Tasks completed:

- Created correlation analysis between lobbying spend and firm ROA with lag structures
- Built visualizations showing relationships across firm size groups and time periods
- Analyzed lag effects to identify optimal timing of lobbying impact (found strongest at 1-2 year lags)
- Cleaned and prepared data for econometric modeling in M3

Hours spent: 10 hours

Key deliverable: Exploratory analysis and lagged correlation findings in capstone_eda.ipynb; summary written in M2_EDA_Summary.md

Milestone 3: Econometric Models (Week 14)

Tasks completed:

- Specified two-way Fixed Effects model with firm and year controls
- Estimated models with clustered standard errors to address heteroskedasticity
- Ran diagnostic tests: Breusch-Pagan heteroskedasticity test, VIF for multicollinearity
- Conducted robustness checks with alternative lags (lag 1, 2, 3) and exclusions

Hours spent: 14 hours

Key deliverable: Fixed Effects regression code and results tables in capstone_models.py; diagnostics and interpretation in results/tables/ and M3_interpretation.md

Milestone 4: Final Investment Memo (Week 14)

Tasks completed:

- Compiled regression tables and model outputs for final memo
- Wrote interpretation of findings for non-technical audience
- Edited sections for clarity and professional tone
- Verified all numbers and citations match results from M3

Hours spent: 8 hours

Key deliverable: Contribution to final memo writing and quality assurance

Total Estimated Contribution

Total hours across all milestones: 44 hours

Percentage of team workload: 25% (equal share of 4-person team)

Role(s) on team: Data analyst and econometrician (M1-M3 focus), supporting M4 with technical validation

2. One Defended Methodological Decision

Decision: I recommended using a 1-year lag for lobbying expenditures in the Fixed Effects model rather than contemporaneous values.

Reasoning:

The M2 exploratory analysis showed that the correlation between lobbying spend and subsequent ROA was stronger at lag 1 (approximately -0.05) compared to contemporaneous values, suggesting a delayed effect. Economically, this makes sense: firms spend money on lobbying in year t, policy decisions take time to implement (averaging 6-12 months), and the financial impact shows up in the following year's accounts. Using lag 1 also reduces endogeneity concerns from reverse causality (firms lobbying in response to current profitability problems).

Data evidence: M2 EDA comparing correlation at lag 0, 1, and 2 showed lag 1 had the most consistent coefficient sign across sample splits and winsorization levels.

Robustness check: M3 robustness table tested lag 1, 2, and 3 specifications; lag 1 model was most statistically usable with fewest data loss issues.

Alternative considered (and why rejected): We considered using contemporaneous lobbying for simplicity and maximum sample size. However, this produced weaker and less economically interpretable results. The lag structure better matches the actual policy-to-finance timeline and improved our ability to interpret the model.

3. One Key Limitation of Our Analysis

Limitation: Our Fixed Effects model assumes that time-invariant firm characteristics (management quality, established relationships in government, core business model) do not bias our lobbying coefficient. However, during COVID-19 (2020), many firms fundamentally restructured their business mix (retail firms exited retail properties, shifted to e-commerce, etc.), which violates this assumption.

Why this matters:

If a firm that lobbies heavily in 2019 then exits a policy-sensitive business line in 2020, our model would attribute the ROA change to lobbying rather than the business restructuring. This can bias our estimate. A firm with stable government relationships might appear less responsive to lobbying if it simultaneously downsizes its exposure to regulated sectors.

Potential mitigation:

A future robustness check could use sector-by-year interactions to allow sector-specific time effects, partially capturing portfolio shifts. Alternatively, we could trim 2020 data entirely or use a dynamic panel approach (Arellano-Bond GMM) that accounts for lagged dependent variables and time-varying heterogeneity. However, this requires longer time series and is beyond our current scope.

4. AI Audit Notes

AI Tools Used:

- GitHub Copilot (integrated in VS Code)

Specific AI Use Example 1: Data Cleaning Function

- Task: Writing function to normalize company names for crosswalk matching
- Prompt: "Write Python function to normalize company names by removing common suffixes and punctuation"
- Output: Template code with regex patterns
- Verification: Tested on sample of company names; manually checked that normalized names matched correctly
- Critique: Initial output didn't handle all edge cases (e.g., "Inc." vs "inc"); I added additional regex patterns and tested again

Specific AI Use Example 2: Diagnostic Test Implementation

- Task: Implementing Breusch-Pagan heteroskedasticity test
- Prompt: "How do I run Breusch-Pagan test in statsmodels and extract p-value"
- Output: Code snippet using het_breuschpagan() function
- Verification: Compared output p-values against manual formula calculation; values matched
- Critique: AI suggested generic implementation; I modified to use clustered errors since our model specifies clusters

Overall AI Use: Approximately 20% of coding (syntax help, debugging, test templates), but all economic reasoning, model decisions, and data interpretation were done independently by the team. All outputs were verified by running on actual data.

5. Self-Reflection

What did I do particularly well?

I was thorough with data validation and quality checks during Milestone 1. I verified that all merges maintained the correct firm-year structure and that no duplicates or missing identifiers slipped through. This foundation made all downstream analysis more reliable.

What could I have improved?

I could have explored time-series methods more deeply in M3. The ARIMA model didn't add much value (same RMSE as naive), and I spent the final week rushing through diagnostics instead of investing more time in alternative specifications that might have been more insightful.

What did I learn from this capstone?

I learned that the hardest part of data science is not the modeling—it's cleaning messy data correctly. I also learned to defend my methodological choices with actual numbers rather than just theory. Using lag 1 instead of lag 0 seems like a small decision, but having EDA evidence behind it made the model much more credible.

6. Attestation

By submitting this individual addendum, I affirm that:

- ✓ All contributions listed above are accurate and honest
- ✓ I have not exaggerated my role or minimized teammates' contributions
- ✓ I understand that this addendum may be used to adjust my individual grade relative to the team grade
- ✓ I take full responsibility for my work and any errors in the sections I authored

Signature:	Alycia Reji	**Date:**	May 1, 2026

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